# Green Dueling Bandits

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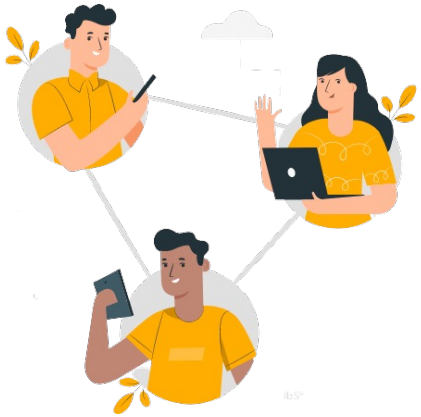
# Green Dueling Bandits: Example in Crowdsourcing

(1) Incoming task/job

Crowdsourcing platform (agent)

(2) Assign task to a pair of workers

(3) Comparison of task's completion qualities (Feedback for dueling)

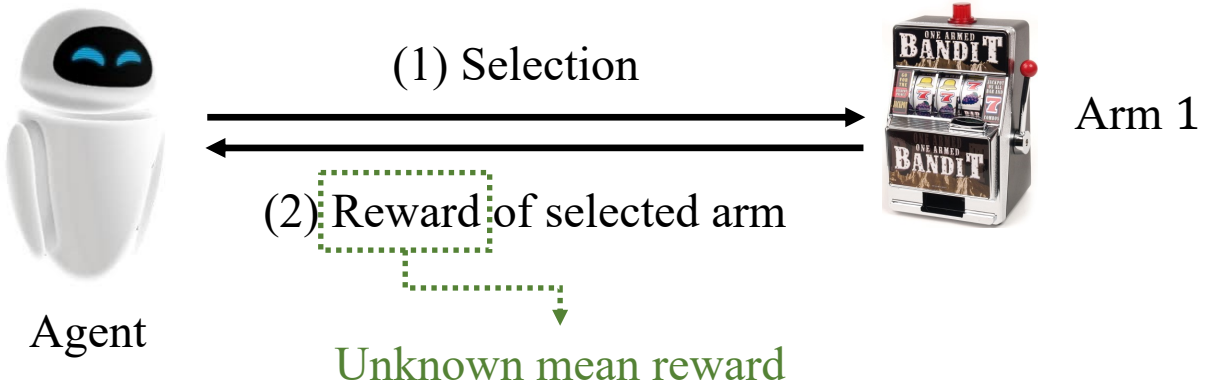


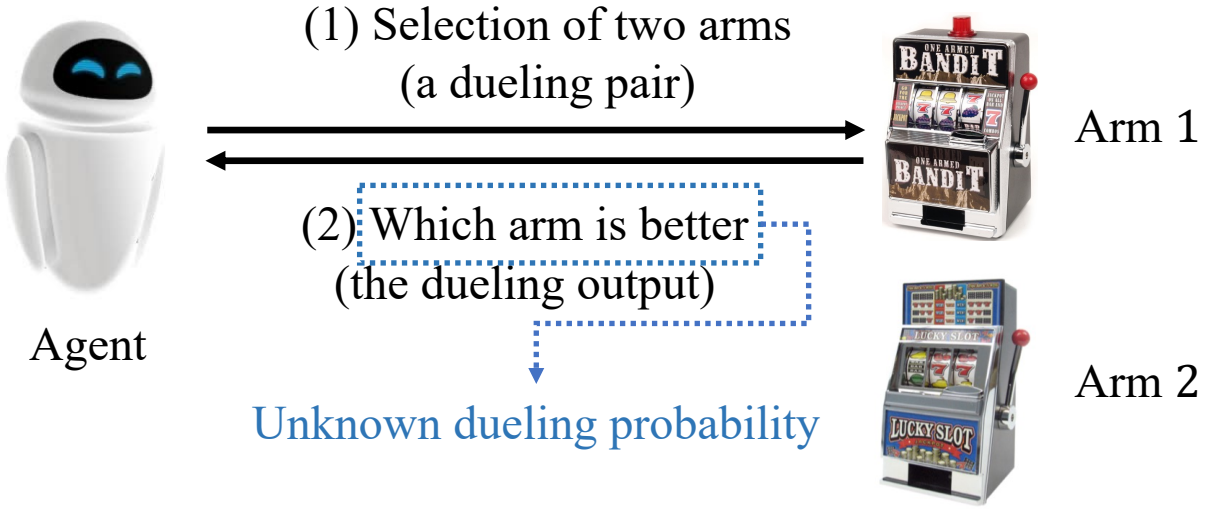
Workers (arms)

- Worker's completion quality is hard and expensive to measure → Pairwise comparison → Dueling bandits
- Limited resources during operation (*e.g.*, time, money, and energy) → Within-budget cost → Green implementation

S. Wang, and Z. Shao. Green Dueling Bandits

# Conventional & Dueling Bandits

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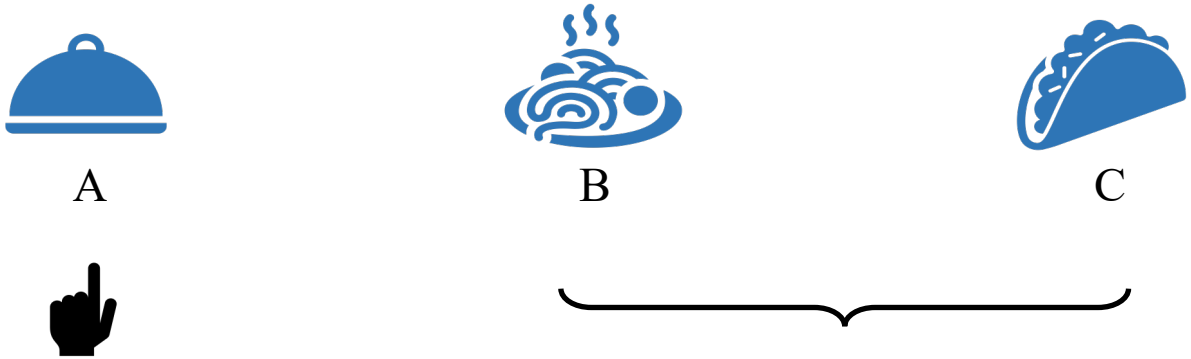
# Pairwise Comparison

Which one is the most delicious?




★ Goodness of taste: 0.8

*(Direct evaluation)*

**Conventional bandits**

★ B tastes better than C

*(Indirect pairwise comparison)*

**Dueling bandits**

No need to explicitly evaluate each dish (arm)

Just need the dishes comparison (dueling of two arms) output

S. Wang, and Z. Shao. Green Dueling Bandits

# Green Implementation & Budget Constraint

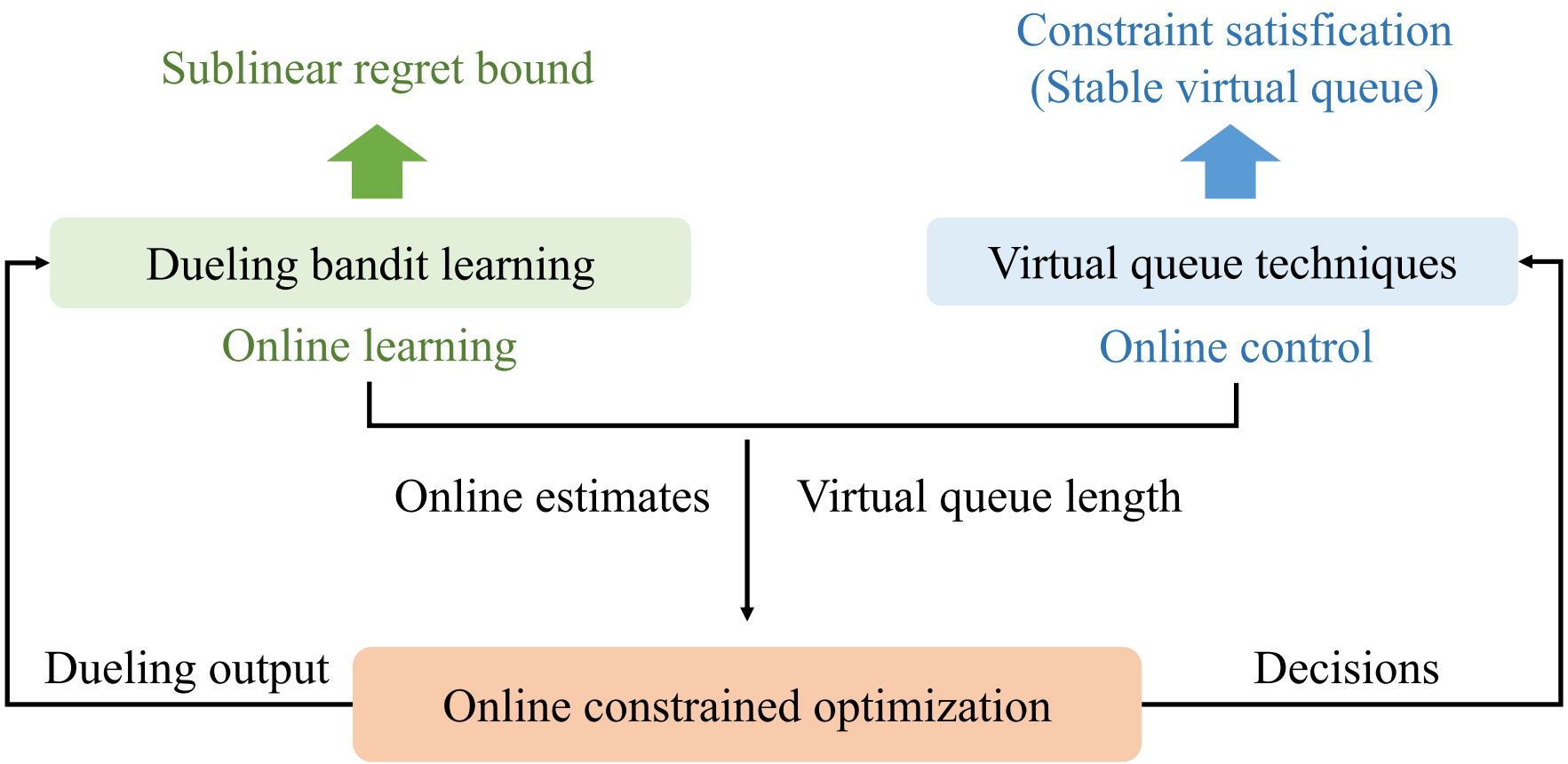
Crowdsourcing platform

Avoid overmuch energy cost for task evaluation

Avoid excessive time cost for task completion

S. Wang, and Z. Shao. Green Dueling Bandits

# Algorithm Design & Key Results




S. Wang, and Z. Shao. Green Dueling Bandits

# Outline

I. Motivation

  I. Indirect Feedback & Dueling Bandits

  II. Green Implementation & Budget Constraint

II. System Model

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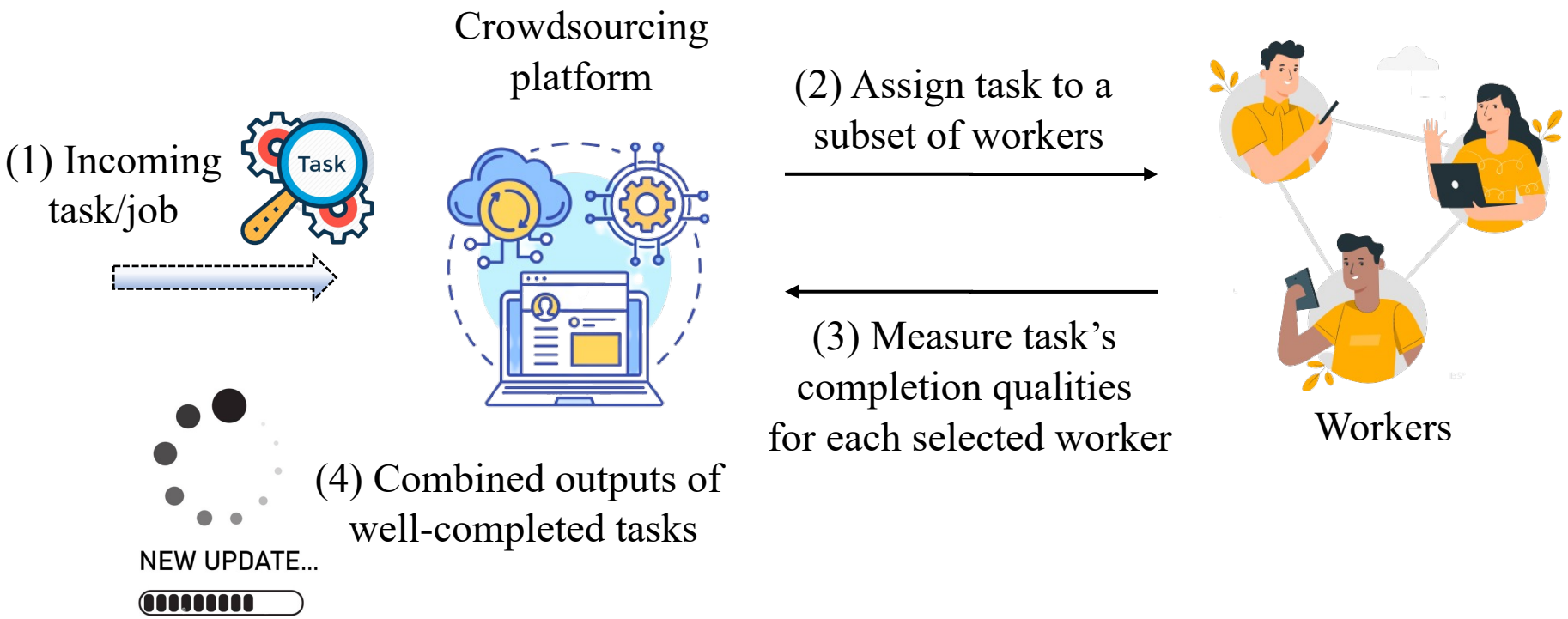
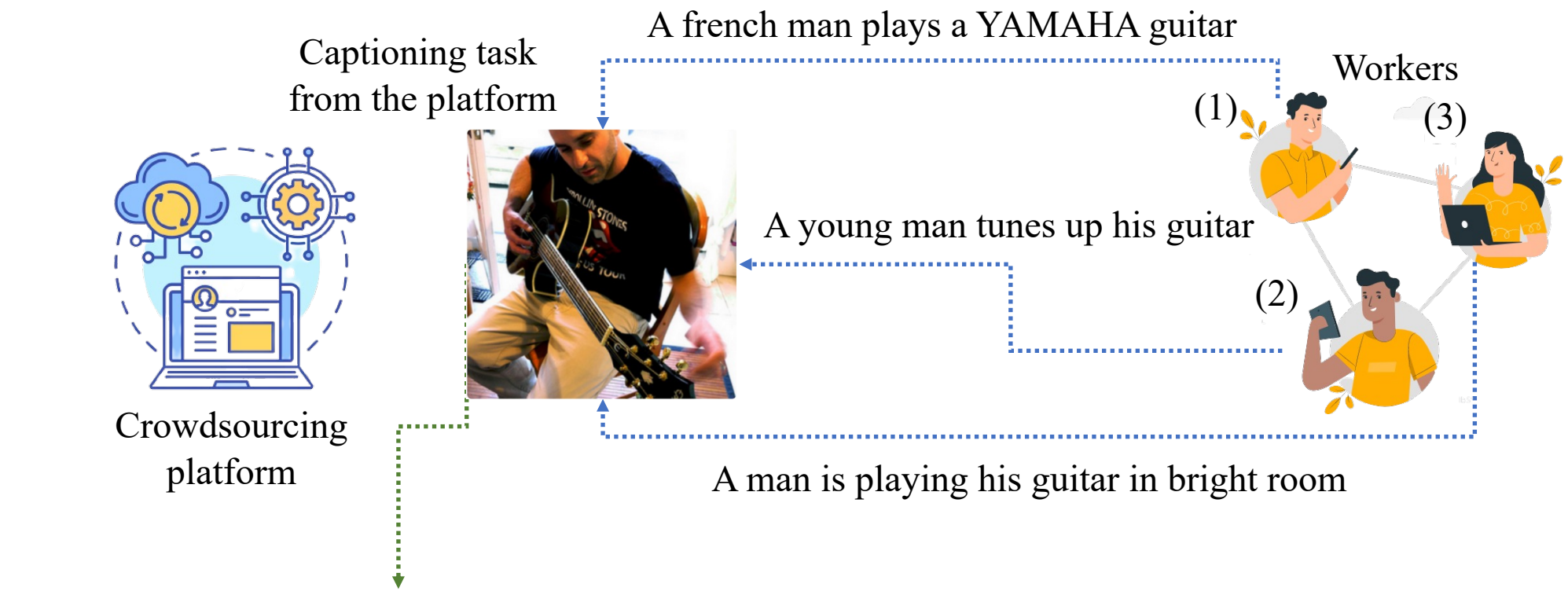
# Task Assignment in Crowdsourcing Systems



# Example: Crowdsourced Image Captioning




*E.g.*, a young man is tuning up his YAMAHA guitar

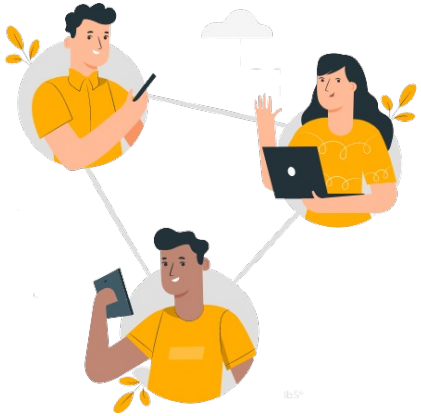
# Unknown Statistics

Worker's unknown average capability for tasks

Crowdsourcing platform

(1) Incoming task/job

(2) Assign task to a subset of workers



Workers

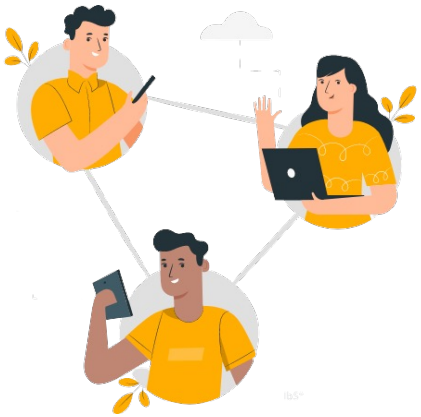
(3) Measure task's completion qualities for each selected worker

(4) Combined outputs of well-completed tasks

# Conventional Online Learning Resolution

Crowdsourcing platform (agent)

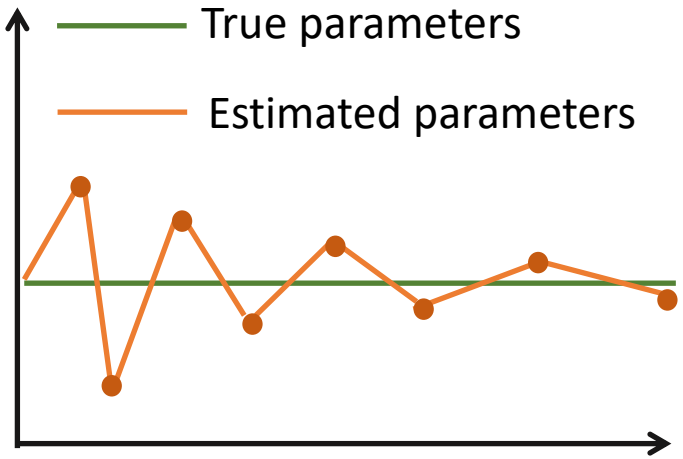
Assign task to a subset of workers



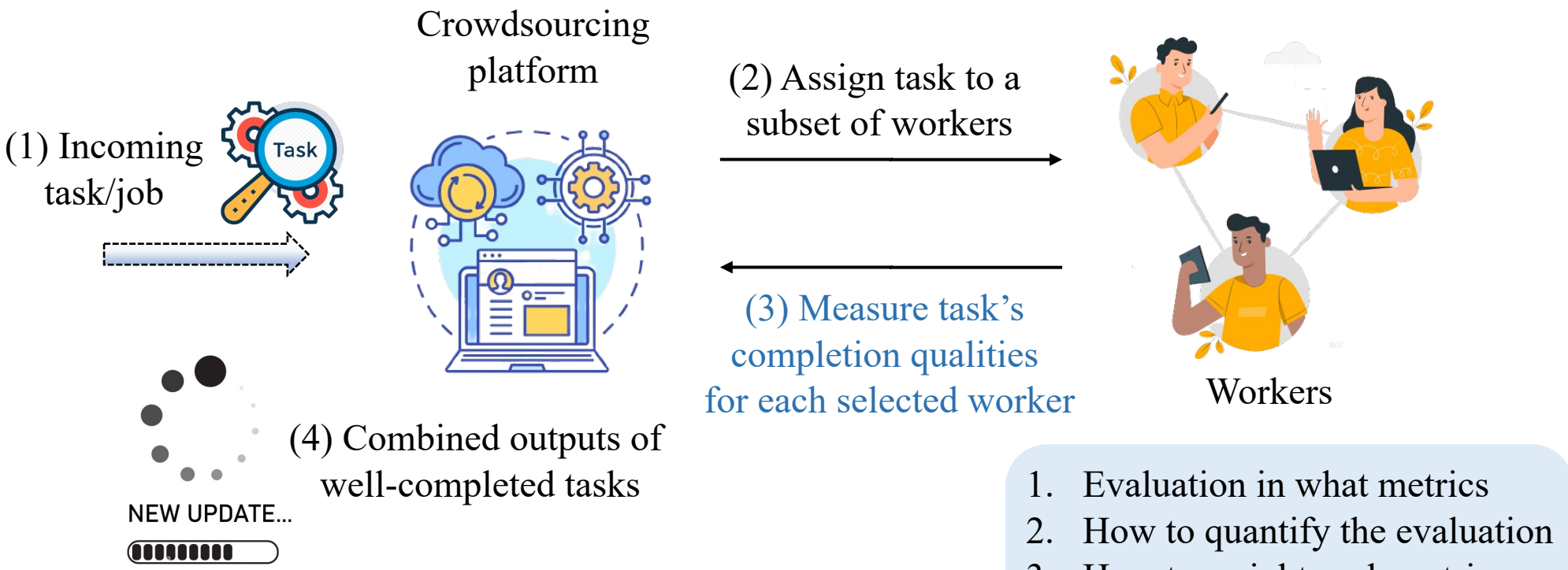
Workers (arms)

Task's completion qualities (reward)

Learning via trial and error



# Key Concern: Hard-to-Evaluate Completion Quality

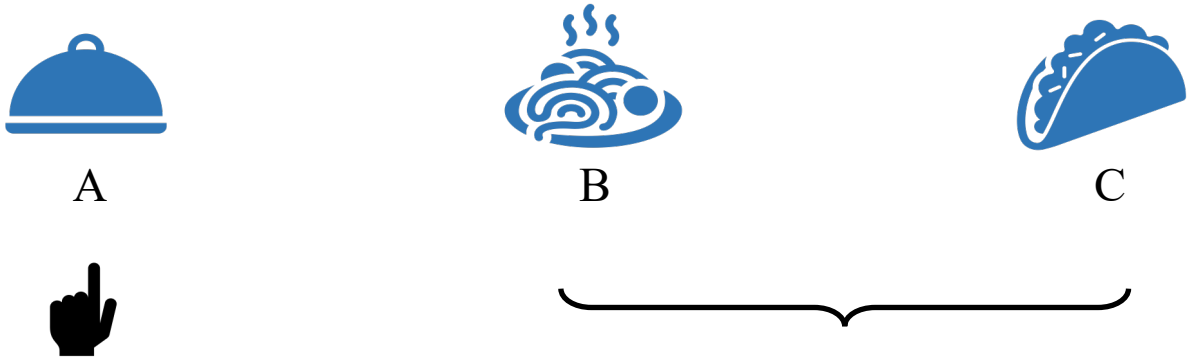



1. Evaluation in what metrics
2. How to quantify the evaluation
3. How to weight each metric
4. What does the evaluation cost

.....

# Indirect Pairwise Comparison

Which one is the most delicious?

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*(Indirect pairwise comparison)*

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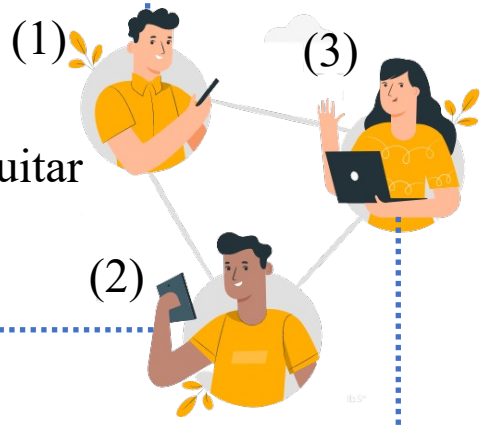
Just need the dishes comparison (dueling of two arms) output

# Orders of Workers (Arms)

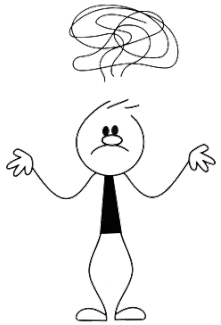
A french man plays a YAMAHA guitar

A young man tunes up his guitar




A man is playing his guitar in bright room

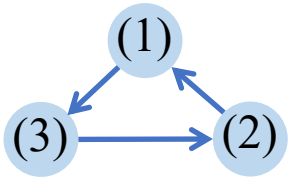
Definition of optimality



Overall concise description

(2) → (1) → (3)

Coherent order




Incoherent order

Precise description about the man and the surrounding

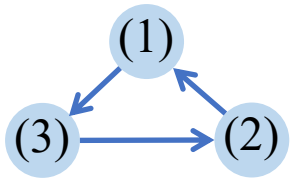
# Borda Score: Metric in an Avergae Sense

Unknown in practice

The Borda score: on average, the prob. that arm (worker) $i$ wins the other $(K-1)$ ones

$$\theta_i = \frac{1}{K-1} \sum_{i \neq j} p_{i,j}$$

The dueling prob. that arm $i$ wins arm $j$




Incoherent order

An aggregated metric of all pairwise comparison

Metric of individual pairwise comparison

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# General Budget Constraint

Crowdsourcing platform

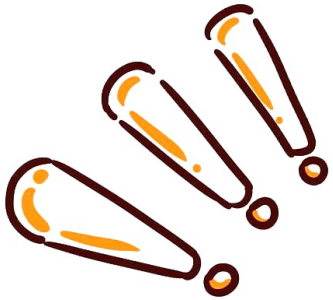
Avoid overmuch energy cost for task evaluation

Avoid excessive time cost for task completion

# Key Observation: Long-term Behavior

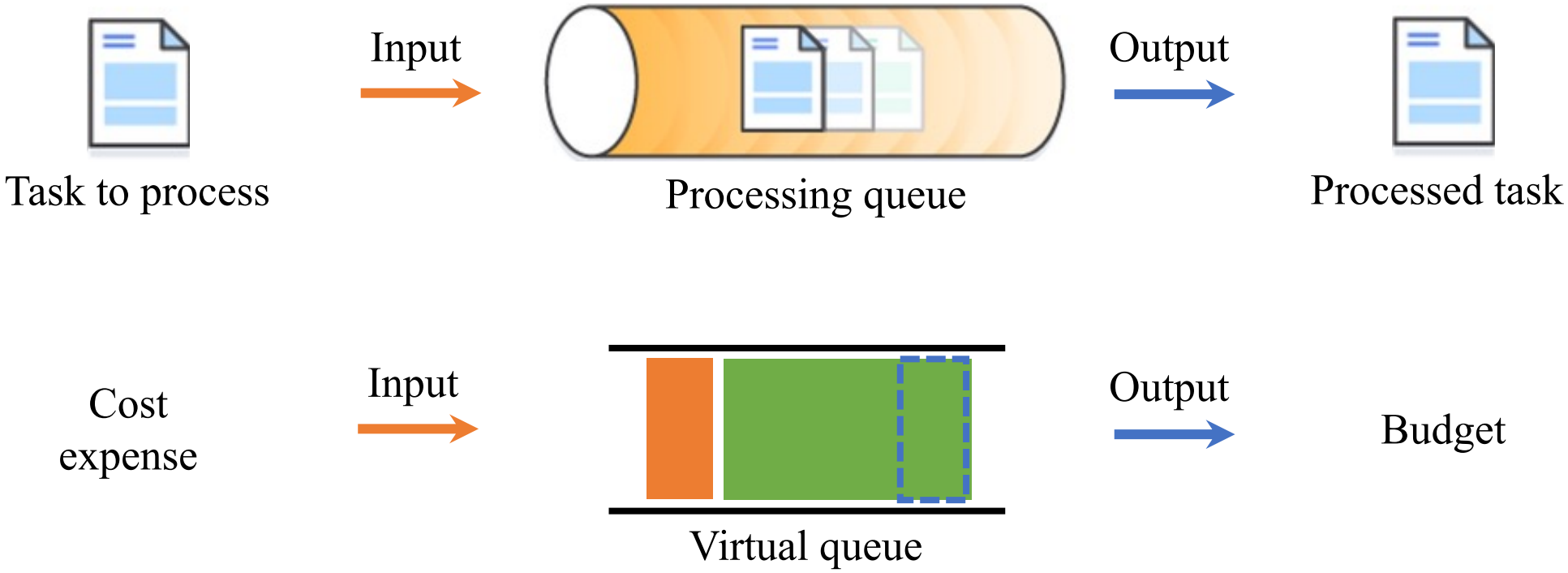
Key observations

- *Long-term deployed* crowdsourcing platform
- *On-going* cost expense and *periodic* allocated budget



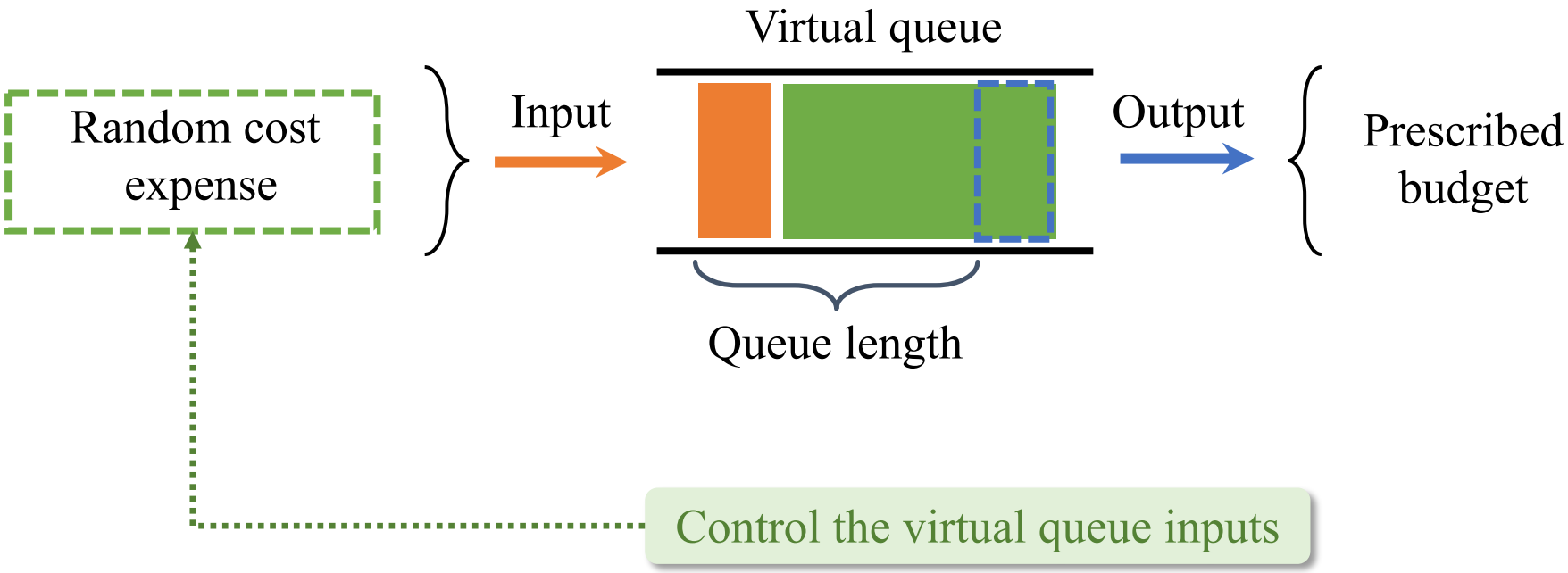
Long-term behavior of cost expense
→ Long-term budget constraint

# Long-term Constraint: a Queueing Perspective



# Constraint Satisfaction: Avoid Overloaded Virtual Queue

# Virtual Queueing with Controlled Inputs



# Key Ingredients Recap

Cost effectiveness

Indirect pairwise comparison

General budget constraint

Dueling bandit learning

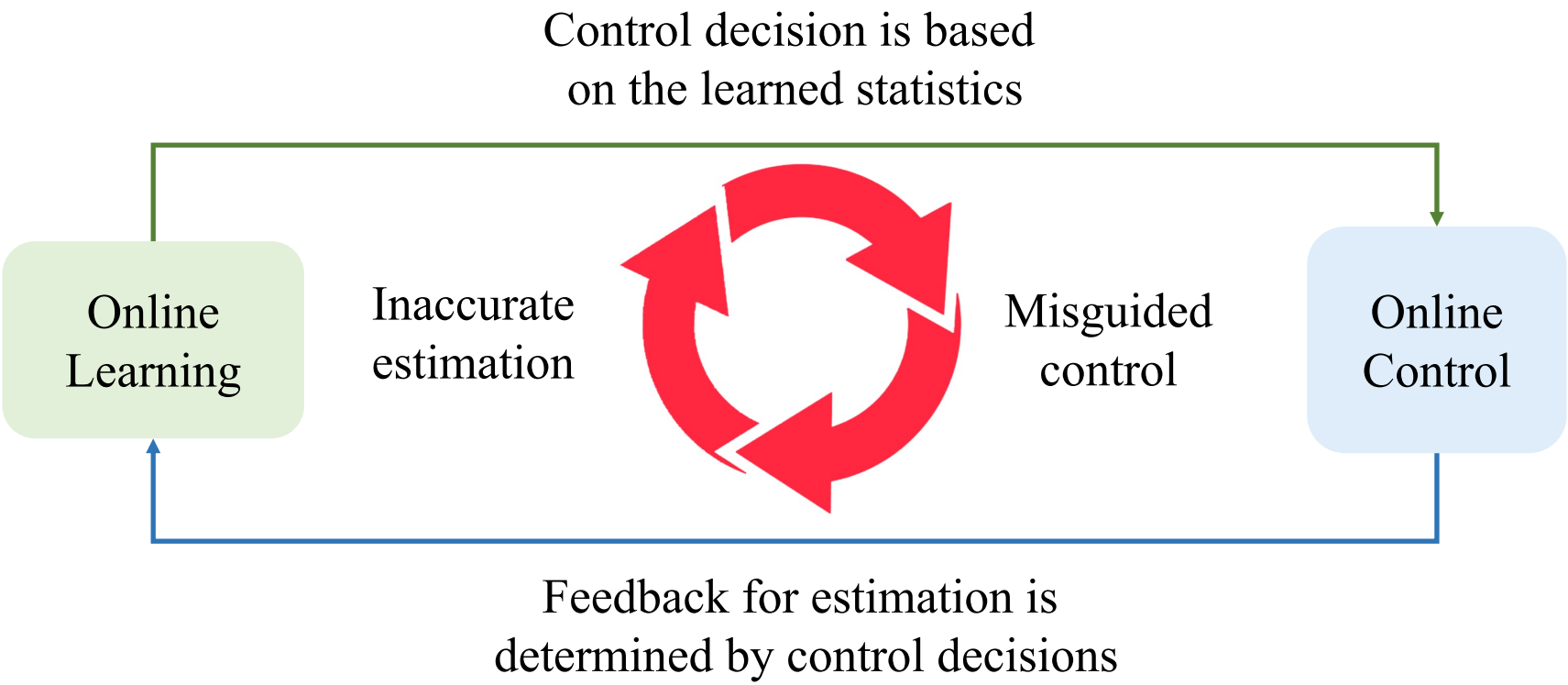
Virtual queueing

Online learning

Online control

# Key Challenge: Integration of Learning and Control

# Coupling of Online Learning and Online Control



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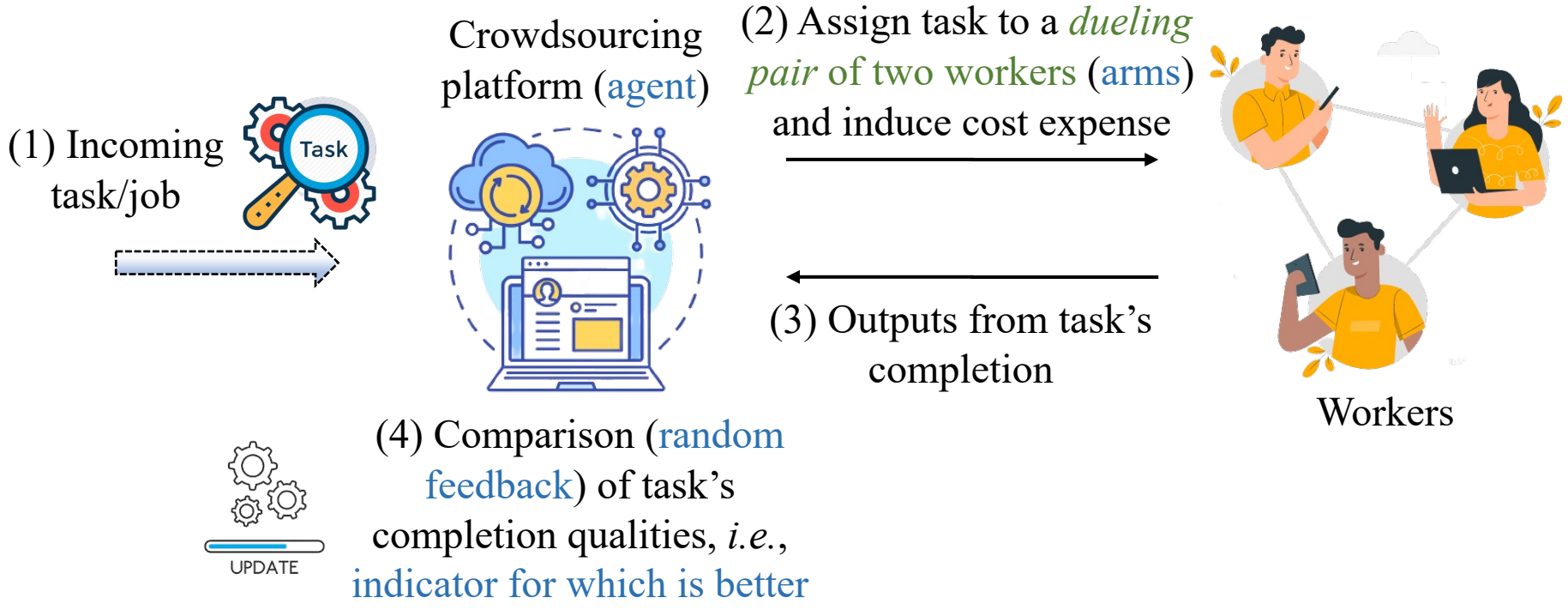
# A Constrained Dueling Bandit Formulation

- Crowdsourcing platform → Agent
- Workers → Arms
- Within-budget cost expense → Long-term budget constraint



# A Constrained Dueling Bandit Formulation

In conventional bandit

- Reward of each arm → Feedback from arm selection

In dueling bandit:

- Reward of each arm → Borda score of each arm

Alternative metric to measure each arm

The feedback is implicit about the quality comparison.
NOT explicit about each (arm) worker's completion quality

*Comparison* (random feedback) of task's completion qualities, *i.e.*, indicator for which is better

# Problem Formulation

Max. the Borda score of arms in the dueling pair = Identidy dueling pair with two arms of highest Borda score ≈ Identidy the best arm via cheap pairwise comparison

$$\underset{\{D(t)\}_t}{\text{maximize}} \ \frac{1}{2} \boxed{\sum_{k \in D(t)} \theta_k}$$

$\theta_k$: the Borda score of arm $k$
$D(t)$: selected dueling pair

$$\text{subject to} \ \limsup_{T' \to \infty} \frac{1}{T'} \sum_{t=1}^{T'} \boxed{\sum_{k \in D(t)} \mathbb{E}\left[C_k(t)\right] \leq b}$$

$C_k(t)$: random cost expense of arm $k$
$b$: prescribed budget for cost expense

Expected cost expense of selected dueling pair is within-budget (in a long-term sense) = Cost expense for task assignment is controlled under budget

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# Bandit Learning with Upper Confidence Bound

The Borda score $\theta_i = \frac{1}{K-1} \sum_{i \neq j} p_{i,j}$ The dueling prob.

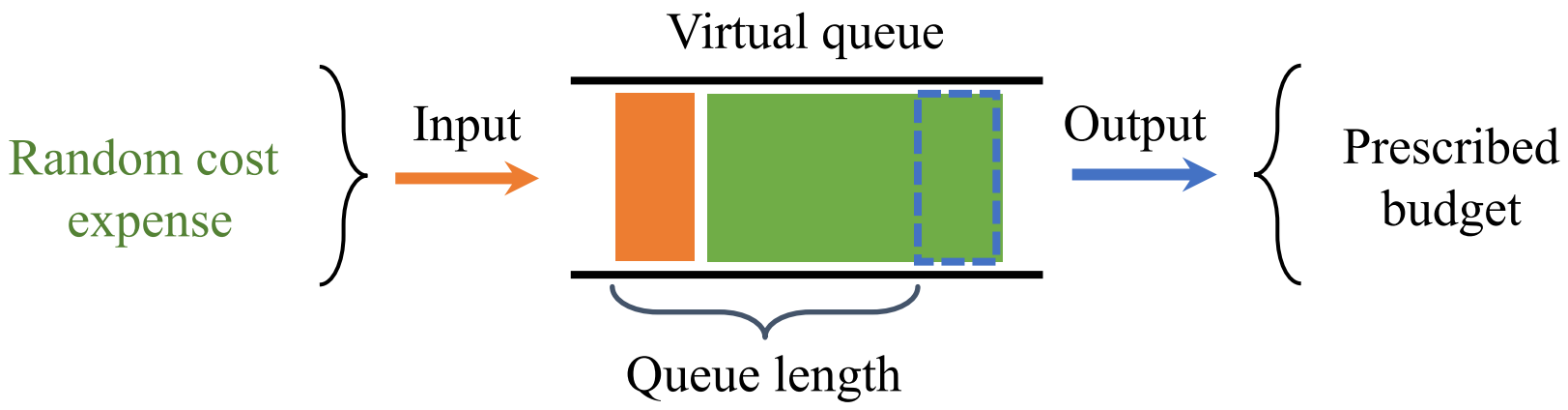
UCB Estimation of *the dueling prob.* = Sample mean + Confidence radius

Exploitation term

Exploration term

Effective bandit learning ⟷ Balanced exploitation-exploration tradeoff

# Online Control with Lyapunov Optimization




Lyapunov optimization

Satisfied long-term constraint ⟷ Stable (non-overloaded) virtual queue

# Integrated Algorithm Design

Online learning

Find best arm (worker)
≈ Maximize Borda score

Max. est. of Borda score
(based on est. of dueling prob.)

Online control

Within-budget cost expense
≈ Minimize queue length

Min. cost weighted
queue length

$$D(t) \in \underset{D \subset [K], |D|=2}{\arg\max} \sum_{k \in D} \frac{V}{2} \widehat{\theta}_k(t) - Q(t) C_k(t)$$

1) Large queue length → select from a *small* set of low-cost arms
2) Small queue length → select from a *large* set involv. high-cost arms

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# Queue Stability

Queue stability definition: $\limsup_{T' \to \infty} \frac{1}{T'} \sum_{t=1}^{T'} \mathbb{E}[Q(t)] < \infty$

Queue length of the virtual queue $Q$ for the budget constraint

Queue stability guarantee: $\limsup_{T' \to \infty} \frac{1}{T'} \sum_{t=1}^{T'} \mathbb{E}[Q(t)] \leq \frac{2H_1 + V}{2\epsilon} < \infty$

Some finite constants

Holds with fininte $V$

# Round-Averaged Regret

Positive tunable param.
to balance regret minimization
and constraint satisfaction

Tunable param. for exploitation-exploration tradeoff with $\alpha > \sqrt{\frac{1}{2}}$

$$\mathrm{Reg}(T) \leq \frac{H_1}{V} + \frac{t_0 + 2H_2}{T} + 2\sqrt{\frac{2\alpha K}{K-1}\frac{\log T}{T}}$$

Some finite constants

Number of arms (*i.e.*, workers)

# Key Observation: Sublinear Regret Guarantee

Online control  Online learning

$$\mathrm{Reg}(T) \leq \frac{H_1}{V} + \frac{t_0 + 2H_2}{T} + 2\sqrt{\frac{2\alpha K}{K-1}\frac{\log T}{T}}$$

Sublinear regret guarantee: $V = \Omega(\sqrt{T/\log T}) \Rightarrow \mathrm{Reg}(T) = O(\sqrt{\log T/T})$

# Key Observation: Scalability

$t_0 = \binom{K}{2}$ rounds to initialize statistics of each dueling pair

$$\mathrm{Reg}(T) \leq \frac{H_1}{V} + \frac{t_0 + 2H_2}{T} + 2\sqrt{\frac{2\alpha K}{K-1}}\frac{\log T}{T}$$

The main term is almost not affected by the scale of number of arms $K$

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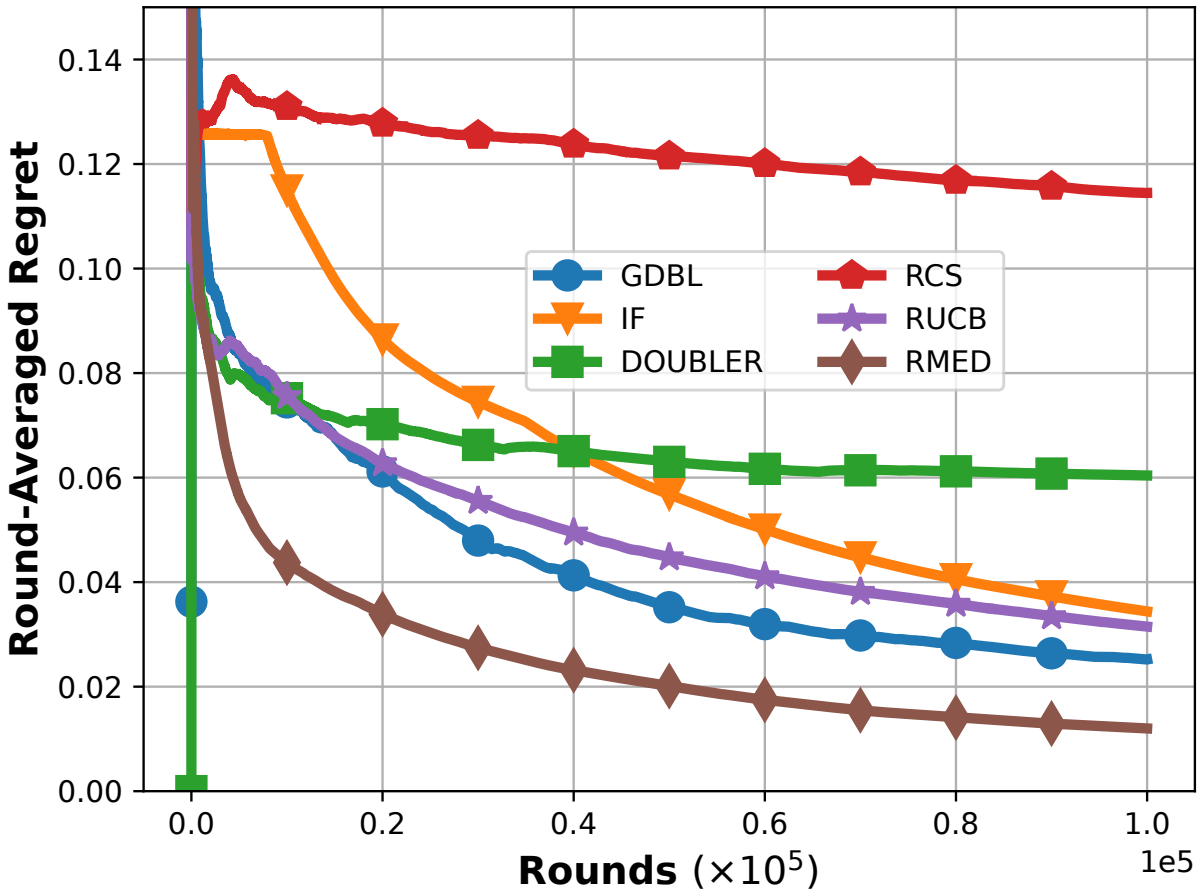
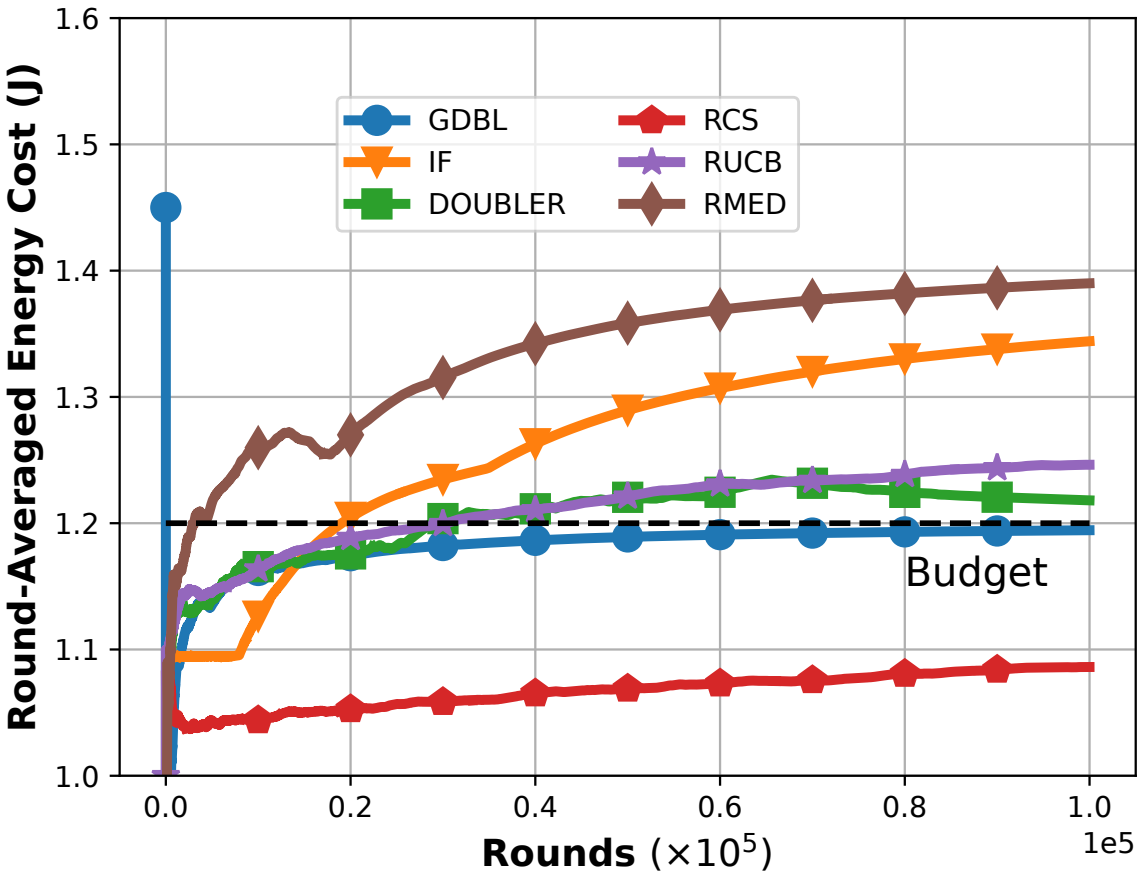
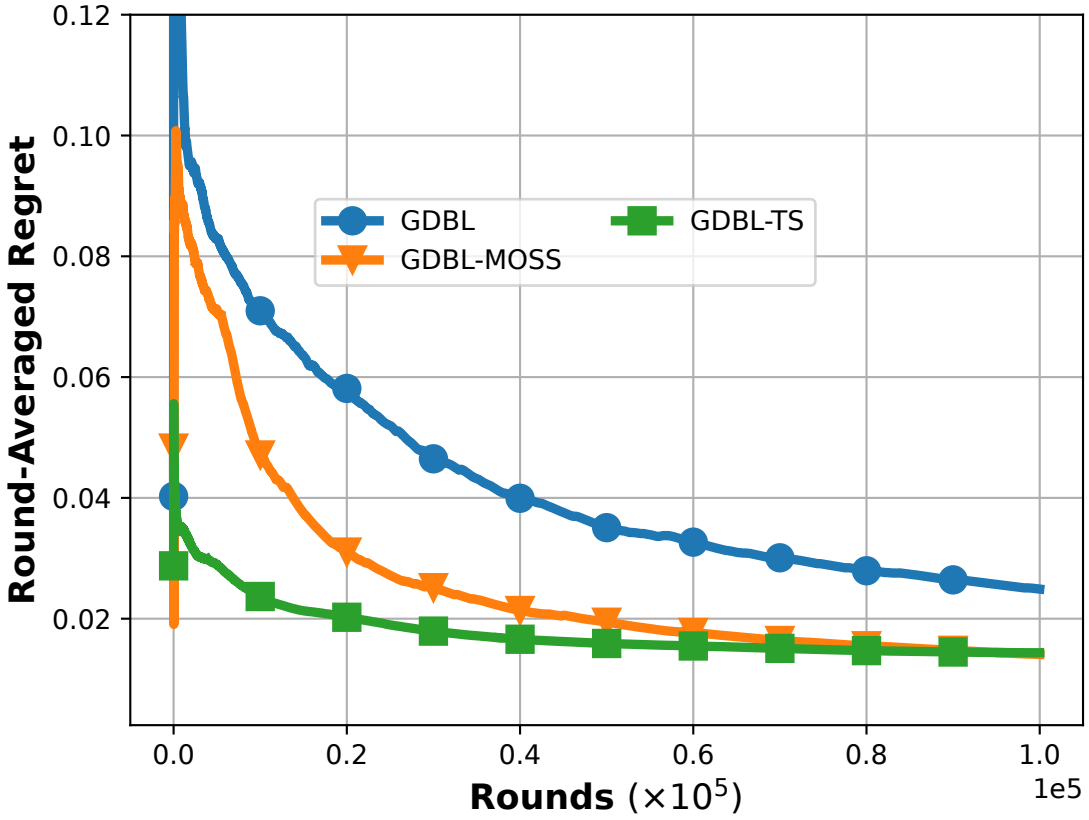
# Simulation Result: Regret



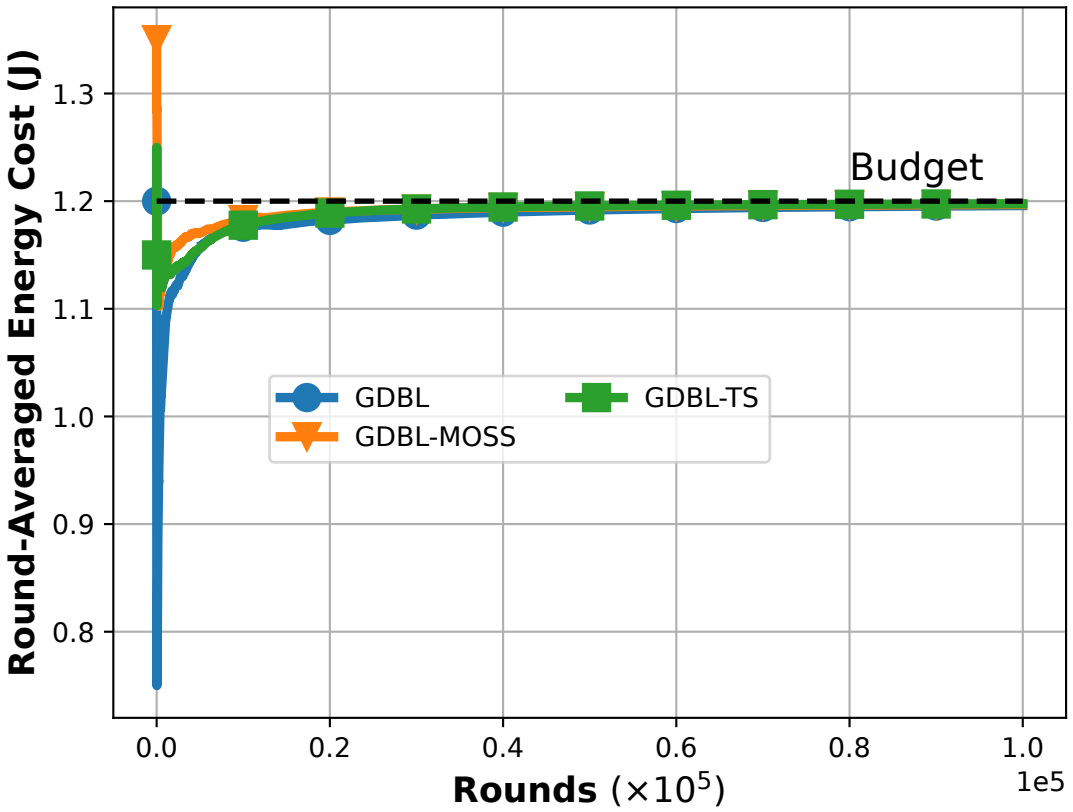
# Simulation Result: Energy Cost



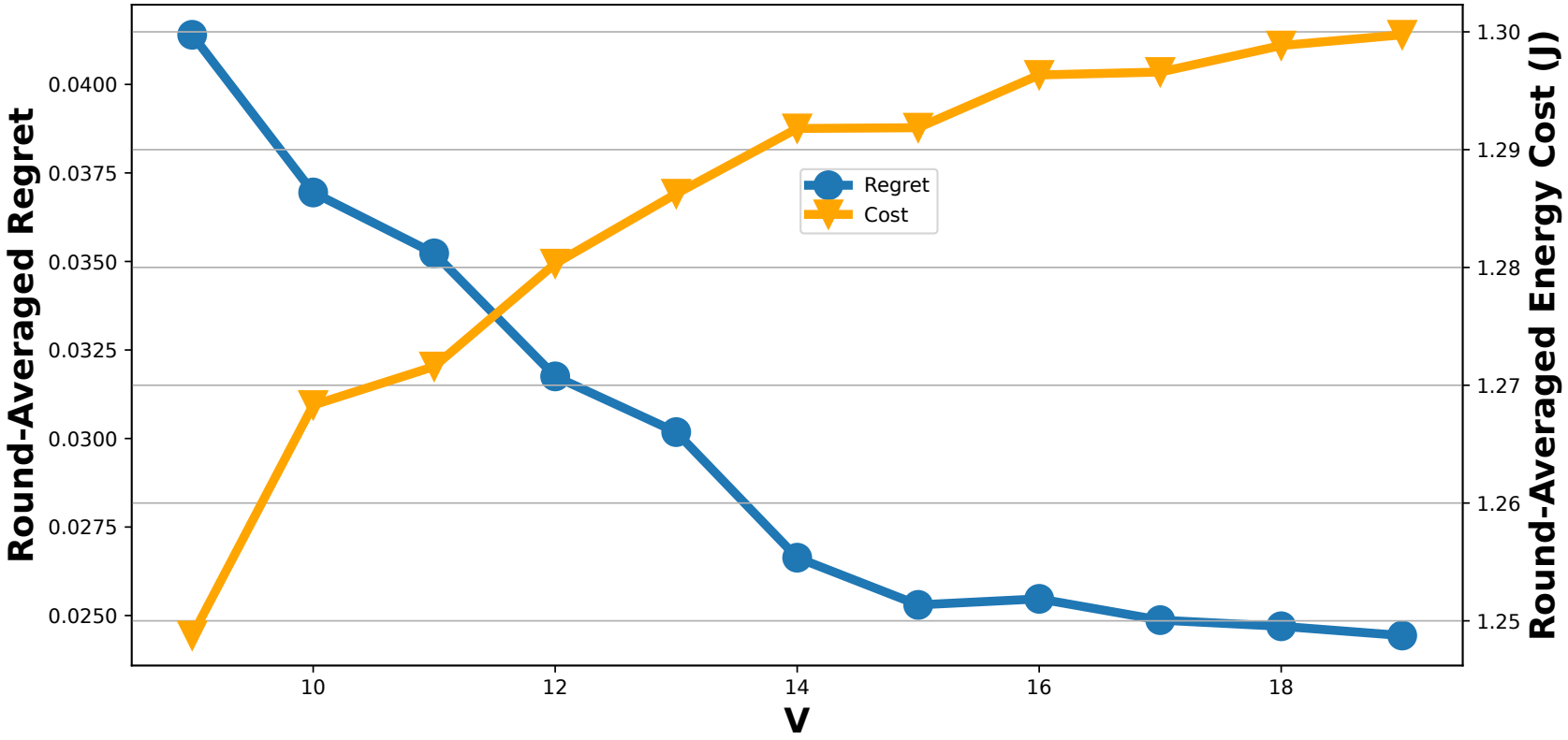
# Simulation Result: Regret of Variants



# Simulation Result: Energy Cost of Variants



# Simulation Result: Regret & Energy Cost



Link to our technical report: http://faculty.sist.shanghaitech.edu.cn/faculty/shaozy/ICC23-GDBL.pdf